DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022

Course: B. Tech. **Branch: Chemical and Petrochemical Engineering** **Semester: VIII**

Subject Code & Name: BTCHC802A Environmental Quality Monitoring & Analysis

Marks: 60 **Date:07/07/2022** **Duration: 3.45 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected to answer as per OBE or the Course Outcome (CO) on which the question is based mentioned in () in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve the following.**		**12**
A)	What do you mean by environment? What is a pollutant? Explain primary pollutants and secondary pollutants with examples.	CO1	6
B)	What is partition constant? Give the nomenclature for the air-soil partition constant of a chemical A.	CO2	2
C)	What is the relation between the partition constant and Henry's Law constant? If the concentration of an organic chemical A in water is 5 mg/L and the equilibrium concentration of A in the air is 2000 mg/m^3. Estimate the air-water partition constant?	CO2	4
Q.2	**Solve any one of the following.**		**12**
A)	What are the physical and chemical parameters of Waste water? State and Explain it.	CO2	12
B)	**i)** The following test results were obtained for a sample taken at the wastewater head works to a waste water treatment plant. All of the tests were performed using a sample size of 50 mL. Determine the concentration of total solids, total volatile solids, suspended solids, volatile suspended solids, total dissolved solids, and total volatile dissolved solids. The samples used in the solids analyses were all either evaporated, dried or ignited to constant weight. **Given Data:** Tare Mass of evaporating dish 53.5433 g. Mass of evaporating dish plus residue after evaporation at 105^{0}C=53.5794g. Mass of evaporating dish plus residue after ignition at 550^0 C = 53.5625 g. Tare mass of Whatman GF/C filter after drying at 105^0 C = 1.5433 g. Mass of Whatman GF/C filter and residue after drying at 105^0 C = 1.5554 g. Mass of Whatman GF/C filter and residue after ignition at 550^{0}C= 1.5476 g.	CO3	6
	ii) What is Solubility? Calculate the solubility of air in water at 0^0 C and 1 atm. pressure. Assume other dissolved material is negligible. Given Data: Henry's constant for air at 0^0 C is 4.32 *10^4 atm/mole fraction.	CO3	6
Q. 3	**Solve Any two of the following.**		**12**
A)	What do you mean by BOD and give its significance? What are the advantages of COD determination over BOD determination?	CO1, CO2	6
B)	Determine the theoretical BOD/COD, BOD/TOC, and TOC/COD ratios for the following compound $C_5H_7NO_2$. Assume the value of the BOD first-order reaction rate constant is 0.23/d (base e) (0.10/d base 10).	CO2	6
C)	Which is a plot of the response of the analytical instrument as a function of the measured variable? Explain. Which are the steps in the sequence of operations in a sampling routine?	CO4	6
Q.4	**Solve the following.**		**12**
A)	In the application of the box model for a segment of a river with the water containing a pollutant, A, which equation is appropriate to estimate the concentration of A in water, when processes occurring in the river are evaporation and reaction of A? Assume no other input or output of water.	CO4	2
B)	For a section of a river, the inlet concentration of a chemical A in water is 50 mg/L at a flow rate of 25 L/s. Using a steady-state box model, estimate the concentration	CO4	4

of A exiting this section. The rate of absorption of A into the water from the air was 5 mg/s, and the rate of reaction of A in the section was 2 mg/s. Report the concentration in mg/L up to 2 decimal places.

C)	What is Sampling? State the names of the Sampling and explain it in short.	CO3	6
Q. 5	**Solve the following.**		**12**
A)	Which of the following dimensionless number represent the ratio of convective mass transport to diffusive mass transport? For the flat plate boundary layer theory for mass transfer from solid surface to air, the correlation is as follows: $N_{SH} = 0.664\ (N_{RE})^{0.8}(N_{SC})^{0.33}$; For laminar flow, $N_{RE} < 5 \times 10^5$ $N_{SH} = 0.036\ (N_{RE})^{0.8}(N_{SC})^{0.33}$; For turbulent flow, $5 \times 10^5 < N_{RE} < 5 \times 10^8$ $N_{SH} = {}^3k_{A1}L/D_{A1}$ Estimate the gas-side soil-air mass transfer coefficient, ${}^3k_{A1}$, for chemical A, in cm/h assuming a fetch of 10m. Relevant properties: Diffusivity of A in air: 6×10^{-6} m^2/s; density of air is 1.3 kg/m^3; viscosity of air is 2×10^{-5} kg/m-s; velocity of air is 1 m/s.	CO5	6
B)	The overall mass transfer flux is a sum of flux by bulk flow and flux by molecular diffusion. Why? explain with the statement. Which of the following resistances is controlling? (a) Air side only (b) Water side only (c) Interfacial (d) Both air and water side	CO5	6

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